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Ch-4 Algebraic Structures.

1. Let $\mathbb{Q}^+$ be the set of all positive rational numbers and $*$ is a binary operation on $\mathbb{Q}^+$ defined by, $a * b = \frac{ab}{3}$ is $\mathbb{Q}^+$ is an abelian group.

Sol^m: Let $a, b, c \in \mathbb{Q}^+$

1. $a * b = \frac{ab}{3}$ is a positive rational number.

$$\therefore a * b \in \mathbb{Q}^+$$

Thus $\mathbb{Q}^+$ is closed

$$\begin{aligned} \therefore (a * b) * c &= \left(\frac{ab}{3}\right) * c \\ &= \left(\frac{\frac{ab}{3} \cdot c}{3}\right) \\ &= \frac{abc}{9} \end{aligned}$$

$$\begin{aligned} a * (b * c) &= \left(\frac{bc}{3}\right) * a \\ &= \left(\frac{\frac{bc}{3} \cdot a}{3}\right) \\ &= \frac{abc}{9} \end{aligned}$$

$$\therefore (a * b) * c = a * (b * c)$$

 $\mathbb{Q}^+$ is associative under $*$.

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3. For $a * e = a$

$$\frac{ae}{3} = a$$

$$\frac{e}{3} = 1$$

 $e = 3$ $e \in \mathbb{Q}^+$ is an identity element

4. For $a * b = e$

$$\frac{ab}{3} = e$$

$$3$$

$$\frac{ab}{3} = 3$$

$$ab = 9$$

$$b = \frac{9}{a}, \text{ since } a \in \mathbb{Q}^+ \text{ \& } a \neq 0$$

 $\therefore b = \frac{9}{a} \in \mathbb{Q}^+$ is an inverse of a

5. For any $a, b \in \mathbb{Q}^+$

$$a * b = \frac{ab}{3}$$

$$b * a = \frac{ba}{3}$$

 $\therefore \mathbb{Q}^+$ is commutative under $*$.

 $(\mathbb{Q}^+, *)$ is an abelian group.

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2. If $G_1 = \{a + b\sqrt{2} : a, b \in \mathbb{Q}\}$, then $(G_1, +)$ is abelian group.

Solⁿ: let x, y be any two elements of G_1 .
Then $x = a_1 + b_1\sqrt{2}$ & $y = a_2 + b_2\sqrt{2}$
where $a_1, a_2, b_1, b_2 \in \mathbb{Q}$

1. closure property:

$$x + y = (a_1 + b_1\sqrt{2}) + (a_2 + b_2\sqrt{2}) \\ = (a_1 + a_2) + (b_1 + b_2)\sqrt{2}$$

if $a_1, a_2 \in \mathbb{Q}$, then $a_1 + a_2 \in \mathbb{Q}$.
also $b_1, b_2 \in \mathbb{Q}$, then $b_1 + b_2 \in \mathbb{Q}$.

Hence $x + y \in G_1, \forall x, y \in \mathbb{Q}$.

$\therefore G_1$ is closed with respect to addition

2. Associative property:

The elements of G_1 are real numbers and the addition of real numbers is associative.

3. Identity element:

$$0 + 0\sqrt{2} \in G_1 \text{ is the identity element of } G_1 \\ \text{as } a + b\sqrt{2} + (0 + 0\sqrt{2}) = (a + 0) + (b + 0)\sqrt{2} \\ = a + b\sqrt{2}$$

$\therefore G_1$ is under the identity element

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4. Existence of inverse:

$$a + b\sqrt{2} \in G_1$$

$$\Rightarrow (-a) + (-b)\sqrt{2} \in \mathbb{Q} \\ \text{since, } a, b \in \mathbb{Q} \\ -a, -b \in \mathbb{Q}.$$

Now,

$$(a + b\sqrt{2}) + (-a - b\sqrt{2}) = (a - a) + (b - b)\sqrt{2} \\ = 0 + 0\sqrt{2} \\ = \text{identity}$$

5. Commutative property

let $x, y \in G_1$

$$x = a_1 + b_1\sqrt{2} \text{ & } y = a_2 + b_2\sqrt{2}$$

$$\therefore x + y = (a_1 + b_1\sqrt{2}) + (a_2 + b_2\sqrt{2}) \\ = (a_1 + a_2) + (b_1 + b_2)\sqrt{2}$$

$$\therefore y + x = (a_2 + b_2\sqrt{2}) + (a_1 + b_1\sqrt{2}) \\ = (a_2 + a_1) + (b_2 + b_1)\sqrt{2}$$

Hence,

$(G_1, +)$ is an abelian group.

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3. $G = \{0, 1, 2, 3, 4, 5, 6, 7\}$ and operation is $+$ addition modulo 8, then $(G, +)$ is an abelian group.

Soln:

$+$	0	1	2	3	4	5	6	7
0	0	1	2	3	4	5	6	7
1	1	2	3	4	5	6	7	0
2	2	3	4	5	6	7	0	1
3	3	4	5	6	7	0	1	2
4	4	5	6	7	0	1	2	3
5	5	6	7	0	1	2	3	4
6	6	7	0	1	2	3	4	5
7	7	0	1	2	3	4	5	6

1. closure property

if $a, b \in G$ $\Rightarrow a + b \in G$, if $a + b < 8$

and if $a + b \geq 8$ then, divide the sum by 8 and the remainder will be less than 8

 $\therefore a + b$ an element of G

2. Associative Property

e.g.,

$$(7 + 2) + 3 = 1 + 3$$

$$= 4$$

$$(2 + 3) + 7 = 5 + 7$$

$$= 4$$

$$\therefore (7 + 2) + 3 = 7 + (2 + 3)$$

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3. Existence of identity

For every $a + b \in G$ $a + 0 \in G$
 $a + 0 = a$

Hence 0 is the identity element for addition modulo 8.

4. Existence of inverse

$$a + b = 0$$

From the table.

Inverse of 0 is 0

Inverse of 1 is 7

Inverse of 2 is 6

Inverse of 3 is 5

Inverse of 4 is 4

Inverse of 5 is 3

Inverse of 6 is 2

Inverse of 7 is 1

Hence every element possesses inverse.

5. Commutative property.

$$a + b = 4 + 3 = 7$$

$$b + a = 3 + 4 = 7$$

Hence G is an additive abelian group

$\therefore \{0, 1, 2, 3, 4, 5, 6, 7, +\}$ is an abelian group of order 8 as G contains 8 elements.

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4. $G = \{1, 2, 3, 4, 5, 6\}$, $X = 'X' \text{ modulo } 7$,
then check (G, X) is an abelian group?

Solⁿ:

$X =$	1	2	3	4	5	6
1	1	2	3	4	5	6
2	2	4	6	1	3	5
3	3	6	2	5	1	4
4	4	1	5	2	6	3
5	5	3	1	6	4	2
6	6	5	4	3	2	1

1. Closure property

From the table,

$$a X b \in G$$

 $\therefore a X b$ an element of G

2. Associative property

$$(3 X 2) X 4 = 6 X 4$$

$$= 3$$

$$(3 X (2 X 4)) = 3 X 1$$

$$= 3$$

3. Existence of identity

For every $a, b \in G$

$$a X e = a$$

$$a X 1 = a$$

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4. Existence of inverse

$$a X b = 1$$

Inverse of 1 is 1

Inverse of 2 is 4

Inverse of 3 is 5

Inverse of 4 is 2

Inverse of 5 is 3

Inverse of 6 is 6

Hence, every element possesses inverse.

5. Commutative property

e.g.,

$$4 X 3 = 5$$

$$3 X 4 = 5$$

Hence, $G = \{1, 2, 3, 4, 5, 6\}$ form an abelian group with respect to multiplication modulo 7.

5. Show that the set $Z_7 = \{0, 1, 2, 3, 4, 5, 6, 7\}$ is a ring under the operations of addition and multiplication modulo 7.

Solⁿ

$$\text{Let } a, b, c \in Z_7$$

(i) $a+b \in Z_7$

as $a+b$ gives the remainder from $a+b$ on dividing by 7

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(ii) Obviously $a + (b + c) = (a + b) + c$

(iii) Since $a + 0 = a = 0 + a$,
0 is the identity element.

(iv) Inverse of 0 is 0, 2 is 7 etc.

(v) Since $a + b = b + a$, + is commutative.

(vi) a.b gives the remainder from 0 to 9
dividing by 9.

$$\therefore a.b \in \mathbb{Z}_9$$

(vii) Obviously $a.(b.c) = (a.b).c$

(viii) Addition + is distributive over under
congruence modulo m.

$\therefore (\mathbb{Z}_9, +, \cdot)$ is a ring

6. Let G be the set of all positive integers
and binary operation $*$ is defined by
 $x * y = xy$. Is G a group.

Solⁿ: Closure property

for every

$x, y \in G$
their product $x * y = xy \in G$
is closed under $*$.

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Associativity property
for every,

$$x, y, z \in G$$

$$(x * y) * z = xyz$$

$$x * (y * z) = xyz$$

$\therefore G$ is associative under $*$.

Identity element

$$x * e = x$$

$$e = 1$$

$\therefore$ The identity element exists and is 1.

Inverse element

$$x * y = e$$

$$x * y = 1$$

If $x > 1$, $1/x$ is a fraction & not a

positive integer.

If $x = 2$, then $y = 1/2$, which is not in G .

$\therefore$ The set of all positive integers with
the binary operation $x * y = xy$ does
not form a group.

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7. show that the algebraic system $(\mathbb{Z}, *)$, where $\mathbb{Z}$ is the set of integers and $*$ is a binary operation defined by $a * b = a + b + 1$ is an abelian group.

Solⁿ: 1. Closure property.

for every $a, b \in \mathbb{Z}$.

$a * b = a + b + 1$ since the sum of two integers is an integer & 1 is an integer, $a + b + 1$ is also an integer.

$\therefore \mathbb{Z}$ is closed under $*$.

2. Associative property

$a, b, c \in \mathbb{Z}$

$$\begin{aligned} a * (b * c) &= a * (b + c + 1) \\ &= a + b + c + 1 + 1 \\ &= a + b + c + 2 \end{aligned}$$

$$\begin{aligned} (a * b) * c &= (a + b + 1) * c \\ &= a + b + c + 1 + 1 \\ &= a + b + c + 2 \end{aligned}$$

$\therefore \mathbb{Z}$ is the operation $*$ is associative

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3. Identity element

$$a * e = a + e + 1 = a \Rightarrow e + 1 = 0$$

$$e = -1$$

$$e * a = e + a + 1 = a \Rightarrow e + 1 = 0$$

$$e = -1$$

$\therefore$, the identity element is $-1 \in \mathbb{Z}$.

4. Inverse element

$$\begin{aligned} a * a^{-1} &= a + a^{-1} + 1 = -1 \\ \Rightarrow a^{-1} &= -1 - a - 1 \\ &= -a - 2 \in \mathbb{Z} \end{aligned}$$

$\therefore$ Every element has an inverse in $\mathbb{Z}$.

5. Commutative

for any $a, b \in \mathbb{Z}$

$$\begin{aligned} a * b &= a + b + 1 \\ b * a &= b + a + 1 \end{aligned}$$

Since $a + b + 1 = b + a + 1$, the operation $*$ is commutative

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8. Let $G = \{1, 2, 3, 4, 5\}$ under the operation of multiplication modulo 6. Write the composition table. Find the inverse of 2, 3 and 5. Is G a cyclic group.

Sol^m:

$\times_6$	1	2	3	4	5
1	1	2	3	4	5
2	2	4	0	2	4
3	3	0	3	0	3
4	4	2	0	4	2
5	5	4	3	3	1

Here, $G = \{1, 2, 3, 4, 5\}$

The element 0 is not in the set G .
 $\therefore G$ is not closed under the given number.

- Inverse of 2, 3 & 5

$$\begin{aligned} 0 \times_6 1 &= 1 \\ 0 \times_6 2 &= 4 \\ 0 \times_6 3 &= 0 \\ 0 \times_6 4 &= 2 \\ 0 \times_6 5 &= 4 \end{aligned}$$

2 does not have an inverse in G .

$$\begin{aligned} 3 \times_6 0 &= 1 \\ \text{From the table} \\ 3 \times_6 1 &= 3 \\ 3 \times_6 2 &= 0 \end{aligned}$$

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$$\begin{aligned} 3 \times_6 3 &= 3 \\ 3 \times_6 4 &= 0 \\ 3 \times_6 5 &= 3 \end{aligned}$$

$\therefore 3$ does not have an inverse in G .

$$\begin{aligned} 5 \times_6 1 &= 1 \\ \text{From the table} \\ 5 \times_6 1 &= 5 \\ 5 \times_6 2 &= 4 \\ 5 \times_6 5 &= 1 \end{aligned}$$

$\therefore$ The inverse of 5 is 5

$\rightarrow$ Here,

$$\begin{aligned} \text{e.g.,} \\ 2 \times_6 3 &= 0 \text{ \& } 3 \times_6 2 = 0 \\ \text{and } 0 \notin G \end{aligned}$$

$\therefore$ This violates the closure property.

As shown above, elements 2 and 3 do not have inverses in G .

$\therefore$ This violates the inverse property.

Since G does not satisfy the group axioms, G is not a group under multiplication modulo 6.

It can't be a cyclic group.

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Q. show that the set $M = \left\{ \begin{bmatrix} a & b \\ 0 & 0 \end{bmatrix} : a, b \in \mathbb{Z} \right\}$ is a ring under the operations of addition and multiplication.

Solⁿ $\Rightarrow$ closure under addition:
for $a, b \in \mathbb{Z}$,

$$\text{for } a+b = \begin{bmatrix} a & b \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} c & d \\ 0 & 0 \end{bmatrix} \\ = \begin{bmatrix} a+c & b+d \\ 0 & 0 \end{bmatrix}$$

Since $a, b, c, d \in \mathbb{Z}$ so $a+c \in \mathbb{Z}$ and $b+d \in \mathbb{Z}$

$\therefore a+b \in M$

$\Rightarrow$ Associativity of addition

let $x, y, z \in M$,

$$(x+y)+z = \begin{bmatrix} a+c & b+d \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} e & f \\ 0 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} a+c+e & b+d+f \\ 0 & 0 \end{bmatrix} \rightarrow \text{ci)$$

$$x+(y+z) = \begin{bmatrix} a & b \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} c+e & d+f \\ 0 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} a+c+e & b+d+f \\ 0 & 0 \end{bmatrix} \rightarrow \text{cii)$$

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from eqⁿ (i) & (ii)

$$(x+y)+z = x+(y+z)$$

$\rightarrow$ Existence of additive identity:

Here,

$$x + 0 = x$$

$$\begin{bmatrix} a & b \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} e_1 & e_2 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} a & b \\ 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} e_1 & e_2 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\therefore e_1 = 0 \text{ \& } e_2 = 0$$

$\rightarrow$ Existence of additive inverse

$$x + y = 0$$

$$\begin{bmatrix} a & b \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} d_1 & d_2 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} d_1 & d_2 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} -a & -b \\ 0 & 0 \end{bmatrix}$$

$$d_1 = -a \text{ \& } d_2 = -b$$

Here,

$$a, b \in \mathbb{Z}, -a \in \mathbb{Z} \text{ and } -b \in \mathbb{Z}$$

$$\therefore x + (-x) = 0_M$$

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⇒ Commutativity of addition:

$$x + y = \begin{bmatrix} a & b \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} c & d \\ 0 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} a+c & b+d \\ 0 & 0 \end{bmatrix} \rightarrow (i)$$

$$y + x = \begin{bmatrix} c & d \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} a & b \\ 0 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} c+a & d+b \\ 0 & 0 \end{bmatrix} \rightarrow (ii)$$

$$\therefore x + y = y + x$$

$(M, +)$ is an abelian group

⇒ Closure under multiplication:

$$x = \begin{bmatrix} a & b \\ 0 & 0 \end{bmatrix} \text{ \& \& } y = \begin{bmatrix} c & d \\ 0 & 0 \end{bmatrix}$$

$$x \cdot y = \begin{bmatrix} a & b \\ 0 & 0 \end{bmatrix} \begin{bmatrix} c & d \\ 0 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} ac & ad \\ 0 & 0 \end{bmatrix}$$

Since $a, b, d \in \mathbb{Z}$, $ac \in \mathbb{Z}$ & $ad \in \mathbb{Z}$

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⇒ Associativity of multiplication:

Matrix multiplication is associative for any $x, y, z \in M$

$$(x \cdot y) \cdot z = x \cdot (y \cdot z)$$

⇒ Left distributivity:

$$x \cdot (y + z) = x \cdot y + x \cdot z$$

$$\begin{bmatrix} a & b \\ 0 & 0 \end{bmatrix} \cdot \begin{bmatrix} c+e & d+f \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} ac & ad \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} ae & bf \\ 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} a \cdot (c+e) & a \cdot (d+f) \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} ac+ae & ad+af \\ 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} ac+ae & ad+af \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} ac+ae & ad+af \\ 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} ac+ae & ad+af \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} ac+ae & ad+af \\ 0 & 0 \end{bmatrix}$$

$$\text{Thus, } x \cdot (y + z) = x \cdot y + x \cdot z$$

⇒ Right distributivity:

for any $x, y, z \in M$

$$(x + y) \cdot z = x \cdot z + y \cdot z$$

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$$\text{for } (x+y).z = \begin{bmatrix} a & b \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} c & d \\ 0 & 0 \end{bmatrix} \cdot z$$

$$= \begin{bmatrix} a+c & b+d \\ 0 & 0 \end{bmatrix} \cdot z$$

$$= \begin{bmatrix} a+c & b+d \\ 0 & 0 \end{bmatrix} \begin{bmatrix} e & f \\ 0 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} a+c & b+d \\ 0 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} ae+ce & af+cf \\ 0 & 0 \end{bmatrix} \rightarrow (ii)$$

$$\text{for } x.z + y.z = \begin{bmatrix} ae & af \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} ce & cf \\ 0 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} ae+ce & af+cf \\ 0 & 0 \end{bmatrix} \rightarrow (iii)$$

Thus, $(x+y)z = x.z + y.z$

Since, all the ring axioms are satisfied,
the set $M = \left\{ \begin{bmatrix} a & b \\ 0 & 0 \end{bmatrix} \mid a, b \in \mathbb{Z} \right\}$

is a ring under the operation of addition
and multiplication.

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10. $\{ \mathbb{Z}_5, +_5, \times_5 \}$ is an example of finite
integral domain which is field.

Soln: $\mathbb{Z}_5 = \{0, 1, 2, 3, 4\}$

$\{ \mathbb{Z}_5, +_5, \times_5 \}$ is an example of finite
integral domain which is a field

$+_5$	0	1	2	3	4
0	0	1	2	3	4
1	1	2	3	4	0
2	2	3	4	0	1
3	3	4	0	1	2
4	4	0	1	2	3

$\times_5$	1	2	3	4
1	1	2	3	4
2	2	4	1	3
3	3	1	4	2
4	4	3	2	1

→ closure property:

$\forall a, b \in \mathbb{F}, a+b \in \mathbb{F}$ and $a \times b \in \mathbb{F}$

Hence $\mathbb{F}$ is closed with respect to addition
and multiplication.

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2. Associative property:

$$\forall a, b, c \in F$$

$$a +_F (b +_F c) = (a +_F b) +_F c$$

$$a \times_F (b \times_F c) = (a \times_F b) \times_F c$$

Hence, '+', 'x' are associative operations only

3. Commutative property:

$$\forall a, b, c \in F$$

$$a +_F b = b +_F a$$

and

$$a \times_F b = b \times_F a$$

Hence '+', 'x' are commutative on F.

4. Distributive property:

$$a \times_F (b +_F c) = a \times_F b +_F a \times_F c$$

Also

$$c +_F (a \times_F b) = c +_F (b \times_F a)$$

Hence, 'x' distributes over addition from left and also from right

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5. Existence of identity:

$$\forall a \in F \exists 0 \in F$$

$$\text{such that } a +_F 0 = a$$

Hence 0 is the additive identity element of F

$$\text{Also } \exists 1 \in F$$

$$\text{such that } a \times_F 1 = a$$

Hence 1 is the multiplicative identity element of F.

6. Existence of inverse:

$$\forall a \in F \exists b \in F$$

$$\text{such that } a +_F b = 0$$

$$b = -a$$

And b is known as additive inverse also
 $\exists c \in F, c \neq 0$

$$a \times_F c = 1$$

$$c = \frac{1}{a}$$

Then c is known as multiplicative inverse

 $(\mathbb{Z}_5, +_5)$ is an abelian grp $(\mathbb{Z}_5 \setminus \{0\}, \times_5)$ is an abelian grp $\mathbb{Z}_5$ is without zero divisors.'x' distributes over '+' from left and also from the right and hence $(\mathbb{Z}_5 \setminus \{0\}, +_5, \times_5)$ is an integral domain which is also field.

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11. Show that $R = \{a + b\sqrt{2} : a, b \in \mathbb{Q}\}$ for the operation $+$, $\times$ is an integral domain but not a field.

Solⁿ: Let $R = \{a + b\sqrt{2} : a, b \in \mathbb{Q}\}$
let $a + b\sqrt{2} \in R$ and $a_1 + b_1\sqrt{2} \in R$

Then,

$$a_1, b_1, a_2, b_2 \in \mathbb{Q}$$

We have

$$(a_1 + b_1\sqrt{2}) + (a_2 + b_2\sqrt{2}) = (a_1 + a_2) + (b_1 + b_2)\sqrt{2}$$

Since $a_1 + a_2 \in \mathbb{Q}$ $b_1 + b_2 \in \mathbb{Q}$

Thus R is closed with respect to addition and multiplication

Further we have $0 + 0\sqrt{2} \in R$ Since $0 \in \mathbb{Q}$ if $a + b\sqrt{2} \in R$, then

$$(0 + 0\sqrt{2}) + (a + b\sqrt{2}) = (0 + a) + (0 + b)\sqrt{2} = a + b\sqrt{2}$$

$\therefore (0 + 0\sqrt{2})$ is the additive identity.

Again if $a + b\sqrt{2} \in R$, then $(-a) + (-b)\sqrt{2} \in R$ and we have

$$(0) + (-b)\sqrt{2} + (a + b\sqrt{2}) = 0 + 0\sqrt{2}$$

$\therefore$ Each element of R possesses additive inverse

$$(1 + 0\sqrt{2}) \times (a + b\sqrt{2}) = a + b\sqrt{2} = (a + b\sqrt{2}) \times (1 + 0\sqrt{2})$$

$\therefore (1 + 0\sqrt{2})$ is the multiplicative identity.

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Thus R is commutative ring with unity

$\Rightarrow R$ is without zero divisors.

Let $(a + b\sqrt{2})$ and $(c + d\sqrt{2})$ be any two elements of R along R .

$$(a + b\sqrt{2}) \times (c + d\sqrt{2}) = (ac + 2bd) + (ad + bc)\sqrt{2}$$

$$a \times c + 2(b \times d) = 0 \text{ \& } (b \times c) + (a \times d) = 0$$

either it happens when

$$a = 0 \text{ \& } b = 0$$

$$c = 0 \text{ \& } d = 0$$

or

$$\text{Thus } (a + b\sqrt{2}) \times (c + d\sqrt{2}) = 0 + 0\sqrt{2}$$

$\Rightarrow$ Either $a + b\sqrt{2} = 0$ or $c + d\sqrt{2} = 0$

$\therefore$ it is an integral domain
But it is not a field.

12. $\{F, +, \cdot\}$ is a field where F is set of all rational nos. and $+$ and $\cdot$ are ordinary addition and multiplication operators.

Solⁿ F is a set of rational nos. and $+$, $\cdot$ are ordinary addition and multiplication.

$+$ closure

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1. Closure property for '+' and '·'.

$$\forall a, b \in F \Rightarrow a+b \in F$$

$$\text{and } a \cdot b \in F$$

Hence F is closed with respect to addition and multiplication.

2. Associative property:

$$\forall a, b, c \in F$$

$$a+(b+c) = (a+b)+c$$

$$\text{and } a \cdot (b \cdot c) = (a \cdot b) \cdot c$$

Hence $+$, $\cdot$ are associative operations.

3. Commutative property

$$\forall a, b, c \in F$$

$$\Rightarrow a+b = b+a$$

$$\text{and } a \cdot b = b \cdot a$$

Hence $+$ and $\cdot$ are commutative on F .

4. Distributive property

$$\forall a, b, c \in F$$

$$a \cdot (b+c) = a \cdot b + a \cdot c$$

$$\text{Also } (a+b) \cdot c = a \cdot c + b \cdot c$$

Hence $\cdot$ distributes over addition from left and also from right.

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5. Existence of identity

$$\forall a \in F \exists 0 \in F$$

$$\text{Such that } a+0 = a$$

hence 0 is the additive identity element of F .

$$\text{Also } \exists 1 \in F$$

$$\text{Such that } a \cdot 1 = a = 1 \cdot a$$

Hence 1 is the multiplicative identity element of F .

6. Existence of inverse

$$\forall a \in F \exists b \in F$$

$$\text{such that } a+b = 0 = b+a$$

$$\text{Then } b = -a$$

And b is known as additive inverse. Also

$$\exists c \in F, c \neq 0$$

$$\text{Such that}$$

$$a \cdot c = 1 = c \cdot a$$

$$\text{then } c = \frac{1}{a}$$

Then c is known as multiplicative inverse.

Hence $(F, +, \cdot)$ is a field.